

Episode B1 | *Demand*

A simple way to organize preferences.

Last time we took a step toward answering the big questions left from Part A.

1. How do we coordinate many trades at the same time?
2. How do we pick an exchange rate when it changes who benefits from the exchanges taking place?
3. And how do we pick where to live on the PPF?

To answer these questions we need to be able to pick between different alternative points on the PPF. We need to think about what people want.

In your recitation you found preferences, quantity demanded, and the full collection of quantity demanded, the individual demand curve. You saw that your actual preferences look a little curved, but we'll approximate this with a linear relationship to make things easy in this class. And you saw that as price goes down, people want to buy MORE.

Like you saw in the simulation, we can ask anyone for their preferences about anything. Amanda-Grace tries to stay healthy by eating foods like spinach and carrots but she also loves chocolate. To get her preferences for chocolate I can ask how many chocolate bars she would be willing to buy at \$4. That's a pretty steep price for just a single bar, so Amanda-Grace tells me zero. She's willing to buy zero chocolate bars at \$4. Amanda-Grace's quantity demanded at \$4 is 0. We can plot this as a point on the price quantity graph.

Show a dot in the color of the points on the graph with a label: individual quantity demanded.

This point represents an ***Individual Quantity Demanded***: the quantity a buyer is willing and able to buy at a given price.

Next I asked Amanda-Grace how many bars she would be willing to buy at \$2. This is a little more reasonable and Amanda likes chocolate, so she tells me her quantity demanded at \$2 is 1 bar. We can plot this point too.

Then at \$1 she wants to buy two. One for now, one for tomorrow.

I can keep going through a range of prices, asking her how many she would buy at each. As the price becomes very low, she decides she would buy many more than she actually wanted to eat in the moment, deciding to give some to her friends in class and hold onto some for later.

I've collected lots of real data like this from real people, all looking very similar in nature.

At each one of these prices, Amanda-Grace's individual quantity demanded represents her willingness and ability to pay at that price. Each point represents a measure of how much she benefits from a chocolate bar. We checked her willingness to pay at only a few prices. In reality though she could have answered these questions at any prices in between. So we'll just connect the points together with lines. This forms something of a jagged curve, what we call her ***Individual Demand Curve***.

Connect her dots and show the definition of Individual Demand Curve.

This has a negative relationship between price and quantity, which tells us that Amanda-Grace will buy more chocolate bars as the price goes down. This is what we call the ***Law of Demand***: a good's quantity demanded falls with its price.

Write out law of demand and its definition

You can also see that individual demand like this is often curved. We could try to approximate this with an equation. But we're trying to keep things simple in this class. So even though we know it's not a perfect match for most people, we're instead only going to use linear Demand relationships.

Show a line through the data points: $P = 2.5 - Q/2$ (adopted 2026-09-13; her dots are $(\$4, 0)$, $(\$2, 1)$, $(\$1, 2)$, $(\$0.50, 4)$, $(\$0.25, 6)$ — the \$2 and \$0.50 answers sit exactly on the line, the \$4 answer sits above it, which is the “your preferences look curved” moment). Then fade out the dots and their connecting lines.

Having an equation for Amanda-Grace’s preferences for chocolate is very handy. It lets us ask lots of questions about how much she likes various things.

Maybe add something here about how to go between the equation for the line and the graph of the line.

We could ask, “How much chocolate will she buy at \$1.5 dollars?”. We never asked her that question directly, so we don’t have any data about it. All we have to do with the equation is plug in 1.5 dollars into the equation to get her quantity demanded.

Plug in 1.5 dollars into the equation, and show a solution set of steps to get quantity demanded.

If it turns out to be a whole number, then we know that’s how many bars she would be willing to buy. And if it’s not a whole number, we could round down to find how many she would buy. Or the decimal would tell us how much she would buy if she were able to buy by weight.

Note: I want to show the first one ending up with a whole number. Then the second one being a decimal. Adopted: $\$1.50 \rightarrow 2$ bars (whole), then $\$1.75 \rightarrow 1.5$ (round down to 1, or 1.5 by weight).

We know the linear relationship doesn’t match her preferences exactly, because it’s linear. If her preferences were actually linear a linear relationship would be great. And if we used a non-linear relationship, we could do exactly the same things there too, although they would be more mathematical work.

We can also ask the question the other way, “How much are you willing to pay for 3 chocolate bars?” This uses the Individual Demand curve to find a price from a quantity. For example, we could plug in a quantity of 3 into the demand curve and get a price of \$1.

Show this on the figure and do the math solution animation.

This price is what we call her Marginal Benefit, how much she would benefit from that third chocolate bar.

Show the definition for Marginal Benefit on screen.

Marginal in this case just means “one more”. So marginal benefit is just a fancy way of saying, what’s the value to you of getting one more. Another way of saying the law of demand, is that her marginal benefit decreases as she gets more chocolate bars. The first item is very nice, the second one is a little less nice, and each additional one is a little less nice than the one before.

Pause for an exercise doing quantity demanded at a price and marginal benefit at a quantity.

Thinking about marginal benefits, if I were to offer a price of \$1, how would Amanda-Grace would respond to this? She has bought no chocolate from me yet. She’s wondering how many she will buy, starting with the first one. Lets zoom in on just the first bar and think about what’s going on there.

Her marginal benefit for her first chocolate bar is \$2.00, which is above the price. She’s willing to pay more than the price. So she’ll buy the first one from me. She spends \$1, which we’ll call expenditure, leaving this difference between her marginal benefit and the price. This difference measures how much *EXTRA VALUE* Amanda-Grace gets from this trade. We call this her Consumer Surplus

Define Consumer Surplus on the scene.

This is one way of measuring how good the trade is on the buyer's side of the trade. There's a parallel measure of goodness on the seller's side too.

So what about the second chocolate bar? Well we think about it in the same way: is Amanda-Grace's marginal benefit higher than the price? If it's not, she won't. If it is, then she'll buy the bar. If she buys the bar, then we find how much extra value she gets from the chocolate bar above what she paid for it.

We keep going until she doesn't want to buy any more. Then we stop. At this point we know how many chocolate bars Amanda Grace bought, her quantity demanded, and we can find her consumer surplus by adding up the difference between each of the bars and her marginal benefit.

*This should look like transparent vertical bars, one per single unit on the horizontal axis, becoming colored if Amanda-Grace decides to buy that unit, with CS shown in one color above, and Expenditure shown in another color below. *

Highlight her quantity demanded on the graph. And keep a running total of her Consumer Surplus.

Measuring consumer surplus this way can work when we have only a few items. But there's an easier way to do it. Instead of thinking about this as vertical bars, we can measure CS as the area under the demand curve that's also above the price line. This forms a triangle.

Show the triangle for CS.

We can find the area of the triangle using geometry you may be familiar with:
 $\text{Area} = \frac{1}{2}hb.$

Show the equation on screen.

We find the height h of the triangle as the difference between the vertical intercept of the demand curve and the price. In this case that's \$1.50.

Show the difference on the vertical axis.

We find the base as the quantity demanded.

Highlight this on the horizontal axis.

Then we plug these numbers into our equation and solve.

Show the numbers going from the graph into the equation, then simplifying the expression.

You can see these two ways of measuring CS give us almost the same number.

Staircase bars (adopted 2026-09-13): bar i 's flat top sits at the line's value at $Q = i$, top-right corner on the line. At price \$1: MBs \$2.00 / \$1.50 / \$1.00, she buys 3 (the third bar is the zero-surplus marginal unit), bar-sum CS \$1.50, expenditure \$3.00; the triangle gives \$2.25 — the gap is the little corners above each step, and it shrinks as units get small.

*Pause here for an exercise question measuring CS. *

Exercise B1 / candidate: give a demand curve, find the quantity demanded at a price, then the consumer surplus at that price. The Fall 2024 Classwork B1 Q1 is the pattern (pumpkin pasties, $P = 17 - Q/6$, CS at 5 and 10 galleons), minus its elasticity question (now B5's). Source: Practice_Bank/Classwork_B1_v3.md. Continuing-cast note: Vignette B1's pasties curve is $P = 12 - Q/2$ — the Exercise could reuse that same curve so recitation and class run one example.

Amada-Grace isn't the only person who likes chocolate. I could run through this exercise for nearly 8 billion people. Lets actually try to get a sense of not just one person's preferences, but the preferences of a whole group.

Market Demand / Simulation: tally the room at a range of prices, building the market curve on the board.

Each of you had your own individual demand curve that I could have found and plotted on the board. What we did here was ask the group for quantity demanded, which is the sum of everyone's individual quantity demanded. This process of adding up everyone's demand gives us what we call the market demand curve.

DECIDED 2026-09-13: 3D from the Sim/ world — MarketRound scaled to ~12 buyers, sellers dropped: agents on the plane walk their bars into the wall, the camera flattens to the (Q,P) plane and the wall resolves into the descending staircase that becomes the market curve. Built as its own step after the 2D episode (a brief placeholder holds the beat until then).

Just like we did with individual demand we can represent the market demand curve with a line. We often just call this the Demand Curve.

Write out the demand equation: $P = 12 - Q/5$ (adopted 2026-09-13) — one spinach demand curve rides from B1 straight into B3's equilibrium; the examples read: at \$5, 35,000 pounds; at \$2, 50,000 pounds.

Similar to the individual demand curve, we can take a price plug it into the demand curve and retrieve the quantity demanded at a given price. At a price of \$5 per pound, we have a quantity demanded of 35,000 pounds per month.

Do the math on screen.

At a price of \$2, we have a quantity demanded of 50,000.

*Do the math on screen. *

Each of these individual bars is the result of a decision to trade or not, with each trade being broken into expenditure and consumer surplus. It would be very difficult to go through each one. So instead we find total expenditure as the number of bars traded times the price.

Show this area.

We can find the consumer surplus as the area of the triangle, which is the area under the demand curve, above the price, done for all things that are exchanged.

Show a line for the height, then run through each of the vertical bars, extending a line for the base as we sweep right through the bars, the areas accumulating as the sweep runs. Then do the math on screen.

Our method for consumer surplus is even more useful here because there is so much to add up.

Closer

Demand as a relationship between my willingness and ability to pay and my quantity demanded is pernicious and complex in some important ways. For example, what if I don't have money? I'm willing but unable to pay. The demand of a rich person will be higher than the demand of a poor person, even if their subjective experience of a chocolate bar is in some sense the same. Keep these limitations in mind.

The demand curve lands us squarely in the ballpark of being able to resolve the question about how we choose the prices that coordinate how much sellers sell and how much buyers buy. Next time we'll build the same set of ideas for the other side of the market: sellers. Then we'll ask what it costs them to say yes to a trade.